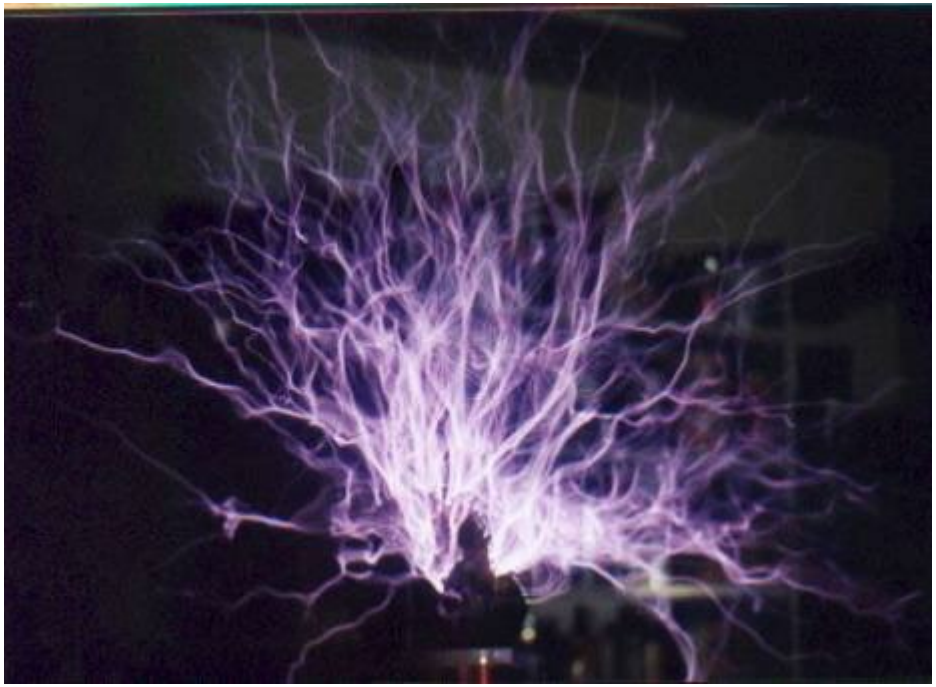
PHYC10009 Foundations of Physics- Lecture 10



Professor Martin Sevior

martines@unimelb.edu.au

Physics, Room 501

(Please switch your mobile phone to silent)

PHYC10009 week3

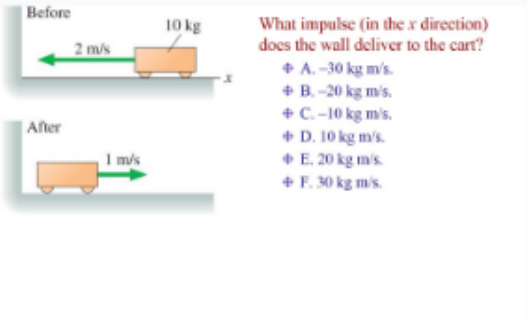
Visual settings

Edit

Respond at **PollEv.com/martinsevior**

Text **MARTINSEVIOR** to **+61 480 025 509** once to join, then **A, B, C, D, E...**

Cart collision



A	**A**
B	**B**
C	**C**
D	**D**
E	**E**
F	**F**

Powered by **Poll Everywhere**

Momentum and Forces.

$$F_{net\ ext} = \frac{dP}{dt}$$

Net force on a system determines its rate of change of linear momentum.

$$(F_{net\ ext})_{av} = \frac{\Delta P}{\Delta t}$$

$$\Delta P = F_{net} \Delta t$$

(Version of the above law for a finite time interval.)

If $F_{net\ ext} = 0$, then $\Delta P = 0$.

Linear momentum of the system is conserved.

Final momentum = initial momentum.

Now Energy… G6

- What is energy?
- Types of energy.
- Energy transformations and transfers.
- The law of conservation of energy.

- Work done by a force.
- Kinetic energy.
- Power.

- Work done by gravity.

Types of energy

Mechanical energy

- Kinetic energy (K)– associated with speed

- Potential energy (U)– energy 'stored' in a system – associated with the configuration of the system (i.e. the relative positions of parts of the system).

Types of energy

Thermal energy

- Associated with the random motions of atoms or molecules.

Energy transformations and transfers – G 6.8

- The total energy of a system is the sum of all the different types of energy in the system.

$$E = K + U_g + U_s + E_{th} + E_{chem} + \ldots$$

- Energy can be *transformed* from one type to another.
  - Energy transformations take place *within* the system.
  - The total energy of the system does not change.

- Energy can be *transferred* into or out of the system.
  - Energy is exchanged between the system and its environment.
  - The total energy of the system may change as a result of energy transfers.

D

Md-2 Wilburforce

V

Wilburforce Video

Law of conservation of energy

If a system is *isolated*, no energy transfers take place to or from its environment.

- i.e. there is no net *external* force on the system *and* there is no heat flow into or out of the system.

Law of conservation of energy:

- The total energy of an isolated system is conserved (has a constant value).

$$E = K + U_g + U_s + E_{th} + E_{chem} + \ldots = \text{constant}$$

- Energy transformations can take place inside the system.
  - These may involve forces acting *within* the system (internal forces).

$$\Delta E = \Delta K + \Delta U_g + \Delta U_s + \Delta E_{th} + \Delta E_{chem} + \ldots = 0$$

$$E_{final} = E_{initial}$$

Law of conservation of energy

If a system is *isolated*, no energy transfers take place to or from its environment.

- i.e. there is no net *external* force on the system *and* there is no heat flow into or out of the system.

Law of conservation of energy:

- The total energy of an isolated system is conserved (has a constant value).

$$E = K + U_g + U_s + E_{th} + E_{chem} + \ldots = \text{constant}$$

- Energy transformations can take place inside the system.
  - These may involve forces acting *within* the system (internal forces).

$$\Delta E = \Delta K + \Delta U_g + \Delta U_s + \Delta E_{th} + \Delta E_{chem} + \ldots = 0$$

$$E_{final} = E_{initial}$$

Kinetic energy - G 6.3

The kinetic energy of an object is:

$$K = \frac{1}{2} m |v|^2$$

- Kinetic energy is a *scalar* quantity.
- It depends on the *magnitude* of the object's velocity (i.e. its speed) but not on the *direction* of motion.

Elastic and inelastic collisions – G6.7

Perfectly elastic collision

- Momentum and kinetic energy are both conserved.
- Objects ‘bounce off’ each other and move separately after the collision.

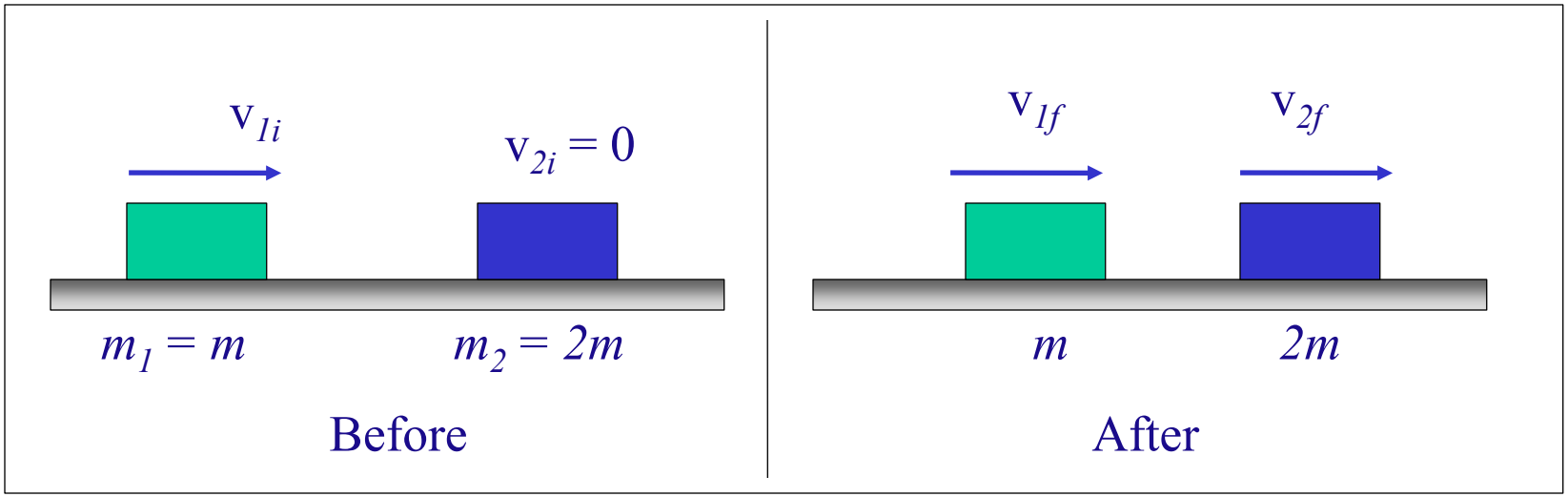
Perfectly inelastic collision

- Momentum is conserved, but kinetic energy is not.
- Objects stick together and move with a common final velocity after the collision.

Many collisions are neither perfectly elastic nor perfectly inelastic.

- Momentum is conserved, but the objects move separately after the collision, as in an elastic collision. Kinetic energy is not conserved.

Example: air track collision - elastic

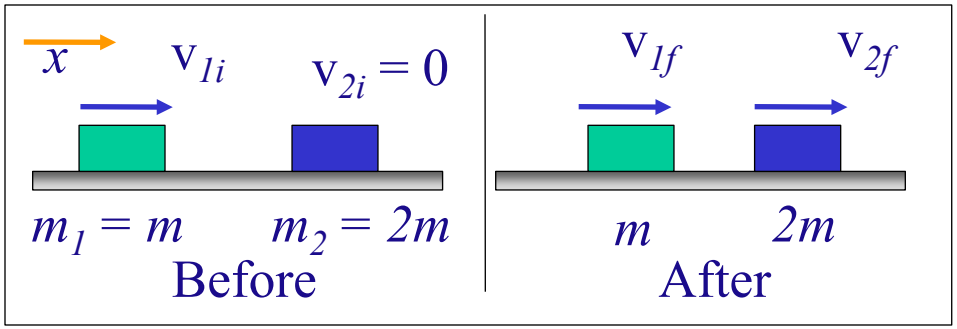


Moving cart hits stationary cart in an elastic collision. Final speeds?

Elastic, so momentum and kinetic energy are both conserved.

We want to find v_{1f} and v_{2f} in terms of v_{1i}.

Example: air track collision - elastic



$P_i = P_f,\ K_i = K_f$

$P = mv_1 + mv_2$

$$K = \frac{1}{2}m_1(v_1)^2 + \frac{1}{2}m_2(v_2)^2$$

$P_i = mv_{1i}$

$P_f = mv_{1f} + (2m)v_{2f}$

$v_{1i} = v_{1f} + 2v_{2f}$

$$K_i = \frac{1}{2}mv_{1i}^2$$

$$K_f = \frac{1}{2}mv_{1f}^2 + \frac{1}{2}(2m)v_{2f}^2$$

$$v_{1i}^2 = v_{1f}^2 + 2v_{2f}^2$$

$v_{1f} = v_{1i} - 2v_{2f}$

$$v_{1i}^2 = (v_{1i} - 2v_{2f})^2 + 2v_{2f}^2$$

$$v_{1i}^2 = (v_{1i}^2 - 4v_{1i}v_{2f} + 4v_{2f}^2) + 2v_{2f}^2$$

$$0 = 6v_{2f}^2 - 4v_{1i}v_{2f}$$

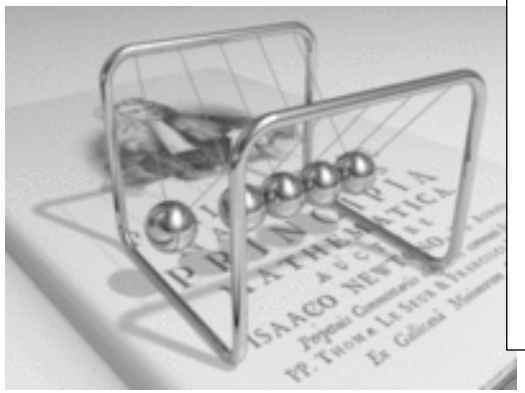
$$2v_{2f}(3v_{2f} - 2v_{1i}) = 0$$

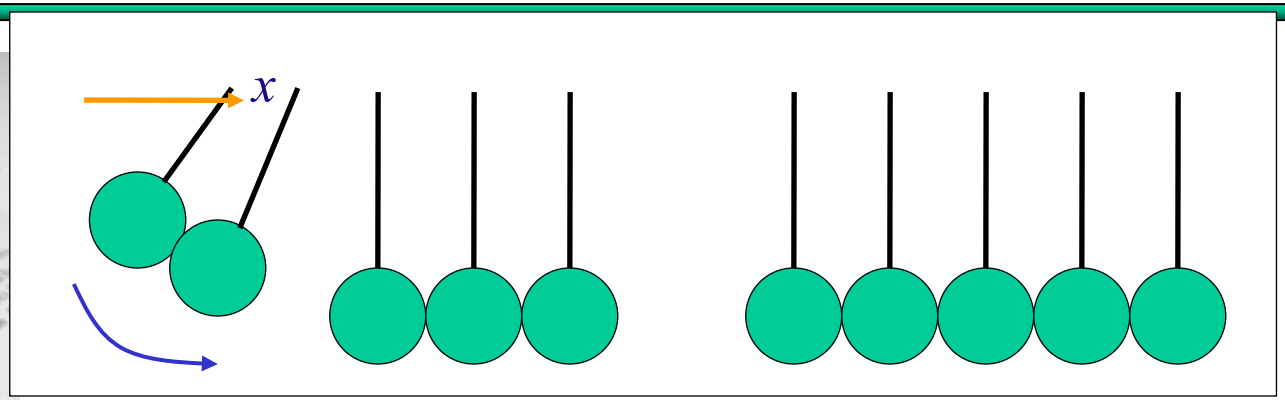
$$v_{2f} = \frac{2}{3}v_{1i} \qquad v_{1f} = -\frac{1}{3}v_{1i}$$

m_1 travels to the left after the collision, at 1/3 its initial speed.

V Mj-6 (Airtrack)

Newton's cradle – G6.7





q balls each has mass *m*. How many leave the right-hand end after the collision (p), and with what speed? (The collision is almost perfectly elastic.)

Conservation of momentum: $pmv_f = qmv_i \quad \Rightarrow v_f = \frac{q}{p} v_i$

Conservation of kinetic energy: $p\left(\frac{1}{2} mv_f^2\right) = q\left(\frac{1}{2} mv_i^2\right) \quad \Longrightarrow v_i^2 = \frac{p}{q} v_f^2$

$$\Rightarrow v_i^2 = \frac{p}{q}\left(\frac{q}{p} v_i\right)^2 \quad \Rightarrow 1 = \frac{q}{p} \quad \Rightarrow p = q \ \ and \ v_f = v_i$$

D Mb-2 Newton's Cradle

V Mb-2 Newton's Cradle

Same number of balls 'out' as 'in', with the same velocity.

If the system is not isolated… - G6.9

- If the system is not isolated, energy transfers in the form of work and/or heat are possible.
- The total energy of the system can increase or decrease.

Change in total energy of the system

$$\Delta E = W + Q$$

Heat transferred into the system

Work done on the system

- This law is a more general form of the law of conservation of energy, known as the *First Law of Thermodynamics.*
- More explicitly:

$$\Delta E = \Delta K + \Delta U_g + \Delta U_s + \Delta E_{th} + \Delta E_{chem} + \ldots = W + Q$$

Mechanical energy – G6.6

- We will focus on the types of energy typically transformed during the motion of ordinary objects – don't worry about chemical energy, nuclear, electrical, etc.
- We restrict ourselves to situations where there is no heat flow (i.e. $Q = 0$) – this is the subject of *thermal physics* (thermodynamics – semester 2).

Total mechanical energy of a system:

$$E_{\text{mech}} = K + U = K + U_{\text{g}} + U_{\text{s}}$$

- How the mechanical energy of system changes, in general:

$$\Delta K + \Delta U + \Delta E_{\text{th}} = W$$

$$(K_f - K_i) + (U_f - U_i) + \Delta E_{\text{th}} = W$$

$$K_i + U_i + W = K_f + U_f + \Delta E_{\text{th}}$$

Kinetic energy

Potential energy

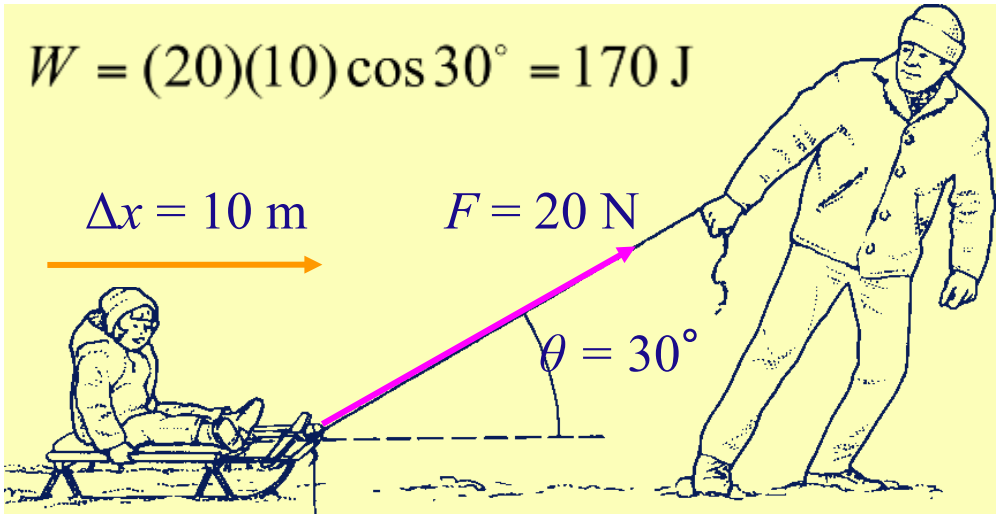
Work done on system by external forces

Thermal energy (produced by friction forces)

Work – G6.1

Work is a transfer of energy to or from a system by the application of a force.

- The system must undergo a *displacement* while the force is being applied for work to be done.



$$W \equiv F_{\parallel}\Delta x$$

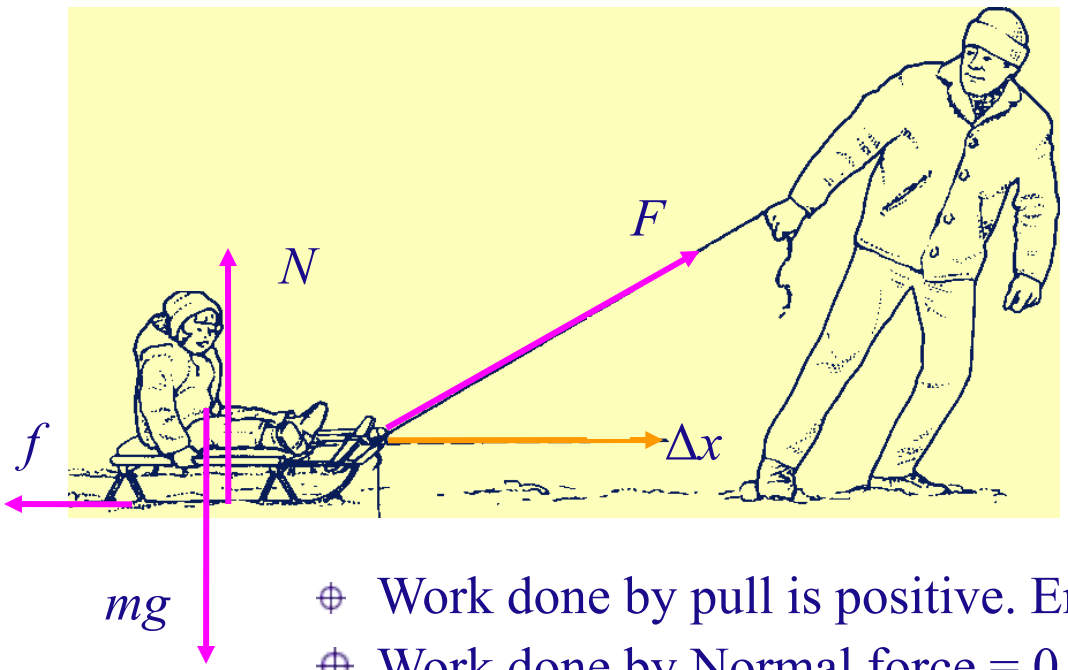
Component of force in direction of displacement

$$W = F(\Delta x)\cos\theta$$

- Work is a *scalar* quantity (it has no direction).
- Units of work: 1 Nm = 1 Joule.
- Work is an energy transfer, so energy has the same units.

Work

Consider the other forces on the *child* + *sled* system.

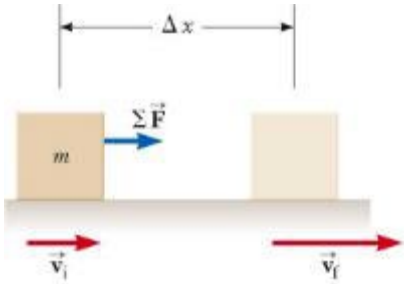


$$W \equiv F_{\parallel}\Delta x$$

$$W = F(\Delta x)\cos\theta$$

- Work done by pull is positive. Energy transferred *to* the sled.
- Work done by Normal force = 0. Work done by weight force = 0.
  - Components of these forces in the direction of motion are zero. ($\theta = 0$).
- Work done by friction is negative. Energy transferred *from* the sled.
  - Component of friction parallel to the displacement is negative.
  - $\theta = 180^\circ$ for the friction force.

Doing net work changes kinetic energy



- The net external force is the vector sum of all forces acting on the system.
- The total (net) work is the sum of the work done by all of the forces acting on the system.
- This example: a constant net force acts as the car has displacement Δx.

$$2a\Delta x = v_f^{\,2} - v_i^{\,2}$$

$$2\frac{F_{net}}{m}\Delta x = v_f^{\,2} - v_i^{\,2}$$

$$F_{net}\Delta x = \frac{1}{2}mv_f^{\,2} - \frac{1}{2}mv_i^{\,2}$$

Doing (net) work causes the system t
speed up or slow down.

$$W_{net} = \Delta K$$

Work-kinetic energy theorem

Example

Olympic Bicycleman rides at a constant speed of 5 m/s. To maintain this speed, he works to overcome forces of friction and air resistance. He applies a constant force of 10 N to the bicycle.

- How much work does he do in riding 100 m?

$$W = F\Delta x = (10)(100) = 1000 \text{ J}.$$

- If he rides this distance in 20 s, at what rate does he do work?

$$\frac{W}{\Delta t} = \frac{1000}{20} = 50 \text{ J/s}.$$

Power – G6.10

Power is the rate of transformation or transfer of energy

$$P \equiv \frac{\text{energy transferred}}{\Delta t}$$

- Units of power: 1 J/s = 1 Watt.
- e.g. Bicycleman's power output = 50 W.

Power due to a force acting on an object travelling at velocity v:

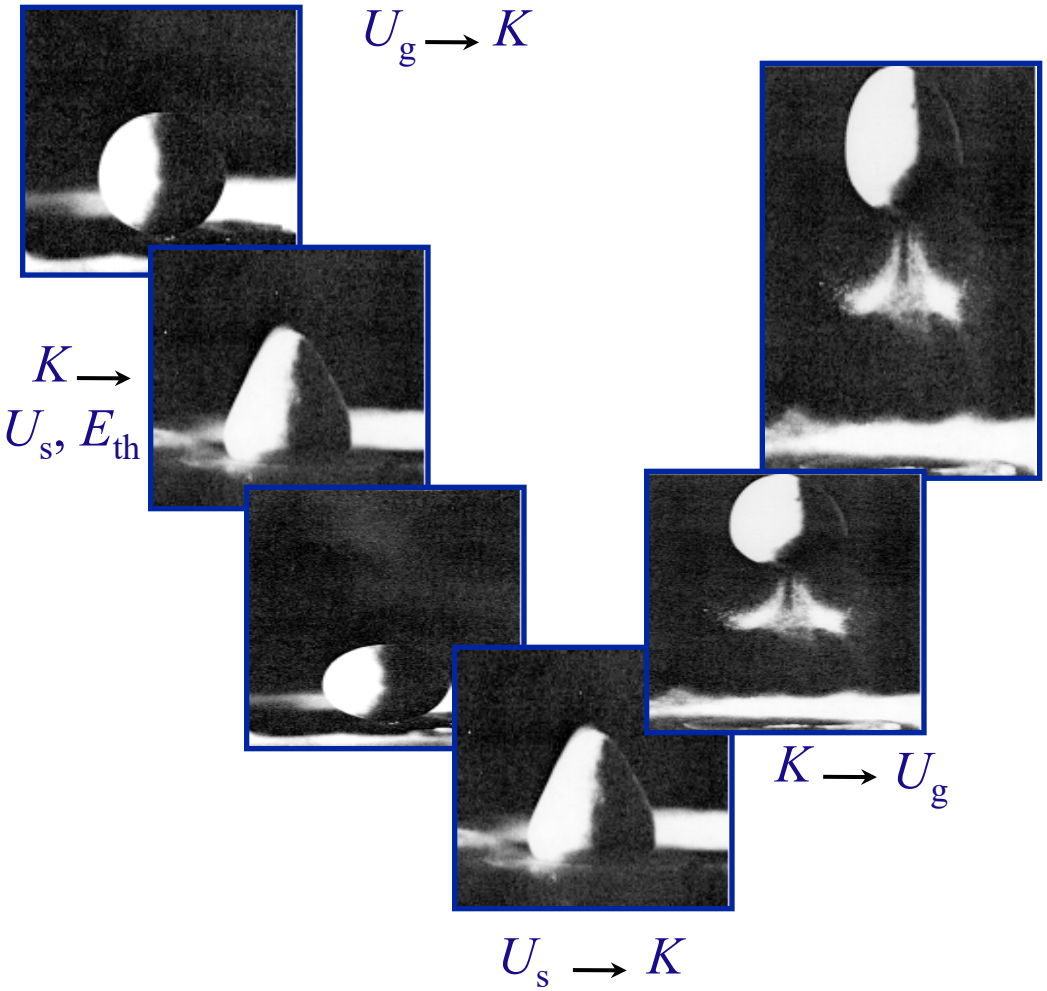
$$P = \frac{F\Delta x}{\Delta t} = F\mathrm{v}$$

- e.g. Bicycleman's power output = (10 N)(5 m/s) = 50 W.

D

Bicycle ergonometer

Example: a bouncing ball



System consisting of ball + Earth (including the ground) is isolated. Total energy is conserved.

$$E = K + U_g + U_s + E_{th} = \text{constant}$$

K = kinetic energy of ball

U_g = gravitational potential energy of Earth-ball system

U_s = elastic potential energy of ball and ground

E_{th} = thermal energy of ball and ground due to random motions of molecules.